

Real FPV Drone Simulator with 3D Scan Data and Visual Effects

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Figure 1: (a) Actual drone piloting at Drone School Tokyo. (b) Third-Person Gameplay Perspective in Drone Simulator. (c) The 3D scanned data of Hino Campus of Tokyo Metropolitan University.

CCS CONCEPTS

• Human-centered computing → Visualization systems and tools.

KEYWORDS

FPV drone simulator, immersion, visual effects, 3D scanning, drone racing

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1 INTRODUCTION

In recent years, FPV drones have seen rapid growth as highly maneuverable and high-speed multicopters, particularly in competitive racing. The demand for FPV drone simulators has also increased,

serving as both flight training tools and gaming products. High-fidelity FPV drone simulators powered by game engines are continuously being released, with *Liftoff* [LuGus Studios 2018], *DCL - The Game* [Drone Champions League, THQ Nordic 2020], and *GlobeXplore* [ozora inc. 2023] being notable examples. However, achieving a truly realistic experience goes beyond sharp video representation. This research focuses on faithfully reproducing optical phenomena such as camera lens aberrations and motion blur, and creating a realistic environment based on 3D scan data. This approach aims to deliver a highly immersive experience that conventional simulators cannot replicate. It is anticipated that this research will offer valuable insights for real-world drone operations, improving situational awareness for pilots and introducing advanced strategic elements to drone racing.

2 METHODS

With assistance from Drone School Tokyo, we received hands-on instruction in actual drone operation, experiencing firsthand the perspective of the onboard camera and the control sensation (Figure 1). The proposed simulator involved multiple workflows, including rectifying mesh and material distortions from 3D scanning. Corrected data was then integrated into a drone simulator with additional visual effects in Unreal Engine. The drone's control sensation is recreated using the Gameready Flyable FPV Drone/Quadrocopter plug-in developed by Vir2alFx [Vir2al-FX 2023]. This simulator can also utilize custom input devices by receiving input from microcontrollers like the ESP32 via serial communication. It is also possible

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to switch between first-person, third-person, and pilot first-person perspectives by pressing keys (Figure 1).

2.1 Preparation of 3D Scanned Data Assets

In a previous study [Nakayasu et al. 2023], a scan of Building 4 of Hino Campus of Tokyo Metropolitan University and part of its courtyard was made using Matterport Pro3 (Figure 1). Mesh distortions were corrected using Blender, and new materials were added to replicate the light sources in the building. Rigid body information was then added in Unreal Engine. The luminous objects' materials were separately configured to adjust their intensity, mimicking actual lighting. Additionally, LED gates, used in drone races, were created in Blender to customize the course.

2.2 Preparation of Virtual Drone

The drone, the primary component of the simulator, was modeled in Maya to replicate a real FPV drone, complete with camera and propeller placements. It was then exported to Unreal Engine in FBX format and associated with the Gameready Flyable FPV Drone/Quadrocopter blueprint [Vir2al-FX 2023]. Additionally, lights were installed on the underside of the drone to mimic those found on actual drones.

2.3 Visual Effects in UnrealEngine

To approximate the real-time reception video of an actual FPV drone, visual effects were simulated in Unreal Engine based on the Elemental Studios tutorial [Elemental Studios 2023]. A clear difference was observed when visual effects were enabled (Figure 2).

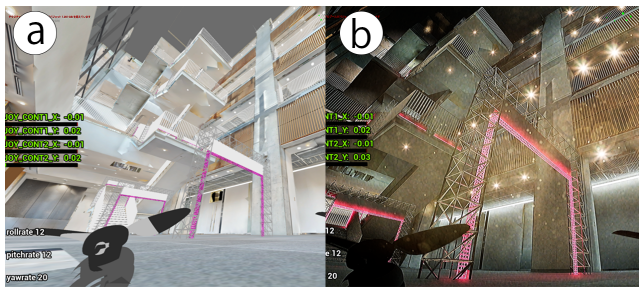


Figure 2: (a) Visual effects absent. (b) Visual effects applied.

3 RESULTS

The drone navigated through the building at high speed under the control of the operator, offering an immersive experience. Both collisions and visual effects were successfully confirmed. As an experiment, we also successfully incorporated high-precision urban spatial data from Japan's Ministry of Land, Infrastructure, Transport and Tourism and Project PLATEAU [MLIT Japan 2024], as well as a USD file of a rocket launch site at JAXA's Tanegashima Space Center. This allowed for a simulated flight experience in areas where drone flights are typically prohibited, thereby enhancing the value of the simulator.(Figure 3)

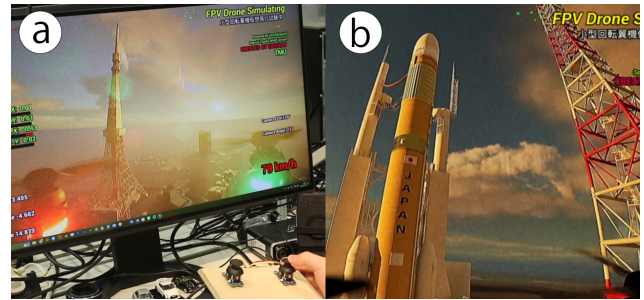


Figure 3: (a) 3D scanned asset from Project PLATEAU is installed and in play. (b) Flight within Tanegashima Space Center.

4 FUTURE WORK

Instances of motion sickness among users were observed, indicating the need for measures such as adjusting the degree of distortion of the wide-angle camera. With the successful integration of USD assets, it is anticipated that NVIDIA Omniverse can be utilized to directly import scene data, including materials and lights, from Digital Content Creation (DCC) tools to Unreal Engine in the future. Additionally, it is expected that new entertainment content will emerge, such as drone races utilizing scan data of buildings restricted due to rights and legal constraints, such as historical landmarks, and virtual tours of important facilities inaccessible to the public.

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